

**Uncertainty-Aware Fusion of Visible and Infrared Imagery for
Reliable Maritime Object Detection**

Progress Report
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Section-A

1. **Project Title** : Uncertainty-Aware Fusion of Visible and Infrared Imagery for Reliable Maritime Object Detection
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Section-B

Progress Report Comprising of Objectives, Methodology, Results, Conclusion and Future Scope

1. Introduction and Problem Statement

Electro-optical (EO) sensing on maritime platforms combines a visible (RGB) camera with a thermal infrared (IR) camera. Each modality fails silently under specific conditions: visible imagery degrades in fog, haze, sun glare, rain and darkness, while infrared imagery sees through glare and darkness but fails during thermal crossover, when a vessel and the surrounding water reach the same temperature. The operational risk is therefore not occasional error but confident error, produced with no accompanying signal that the sensor has become unreliable.

Two gaps motivate this work. First, almost all published maritime detectors output no uncertainty at all. Second, existing visible-infrared fusion is static: it blends the two streams on a fixed rule or a learned-but-static attention map, never conditioned on a live per-frame estimate of how reliable each sensor currently is. This project sits in that gap. It develops a detector that, in a single forward pass, also estimates how much its own prediction can be trusted at that instant, and uses that self-assessed reliability to decide, frame by frame, how much weight to place on each camera.

2. Objectives

O1 — Baseline. Establish a deterministic maritime detection baseline by selecting a YOLO backbone through a controlled, multi-seed benchmark.

O2 — Aleatoric uncertainty (core contribution). A single-pass probabilistic detection head that predicts a variance for every box coordinate, trained with a negative log-likelihood objective, giving per-detection localization uncertainty at real-time speed.

O3 — Distributional / epistemic uncertainty. A low-cost frame-level estimator based on feature-space distance to the training distribution, to catch whole-sensor degradation that per-detection uncertainty misses.

O4 — Uncertainty-gated fusion. A per-modality reliability score derived from O2 and O3, used to adaptively fuse the visible and infrared streams so that the more reliable modality dominates instant by instant.

O5 — Validation. Evaluate calibration — whether predicted uncertainty genuinely tracks real error — and demonstrate that uncertainty-gated fusion outperforms single-modality detection and uncertainty-blind fusion under adverse conditions.

3. Methodology

3.1 System architecture

The detector is held fixed apart from the uncertainty machinery, so that any measured effect is attributable to the uncertainty method rather than to a stronger detector. The aleatoric head is implemented as an in-place conversion of the Ultralytics detection head rather than a forked codebase: a log-variance branch is added alongside the existing distribution-focal-loss (DFL) branch, parameterised over the same left-top-right-bottom distance targets in stride units and converted to pixels at inference. The variance branch reads detached features, so the deterministic detector trains identically to the baseline by construction and the two can be compared fairly. A warm-up schedule holds the likelihood weight at zero for the first epochs while the detector stabilises. A second, cost-free uncertainty signal is read from the spread of the DFL bin distribution of the same trained model, giving a same-model ablation with no additional training.

Frame-level uncertainty uses a Mahalanobis distance in pooled feature space, fitted on clean validation data. The two signals are combined into a per-modality reliability score, which weights a modified weighted-box-fusion step that merges the visible and infrared detections in a common image plane obtained from the dataset calibration. Monte-Carlo Dropout and Deep Ensembles are implemented as reference baselines for the comparison required by O5.

3.2 Data and preprocessing

The primary testbed is the Pohang Canal Dataset with PoLaRIS bounding-box annotations (two classes: ship and buoy), which provides co-registered stereo visible imagery at 2048×1080 and thermal infrared at 640×512, both at 10 Hz, across five runs, one of which is recorded at night. A second dataset, the MIT Sea Grant AUV Lab marine perception dataset, is being annotated in parallel for cross-dataset validation.

3.3 Benchmarking protocol

The backbone selection benchmark was pre-registered before any run: a fixed training recipe (100 epochs, 640×640 input, identical augmentation and optimiser settings) applied unchanged to every candidate variant across the YOLOv8, YOLOv9, YOLOv10, YOLOv11 and

YOLO12 families and their size ladders, each trained with three independent seeds. The selection rule was also fixed in advance — highest mean mAP@50-95, with ties inside one standard deviation broken by DFL compatibility, then integration simplicity, then inference speed — so that the choice of backbone could not be made after seeing the results.

4. Work Completed and Results

4.1 Architecture planning and implementation — complete

The full system architecture has been designed, reviewed and implemented. All components required by objectives O1 to O5 exist in code: the Gaussian variance head and its training loop, the DFL-derived uncertainty extractor, the Mahalanobis out-of-distribution scorer, the reliability computation, the reliability-weighted fusion stage with its coordinate transform, the Monte-Carlo Dropout and Deep Ensemble baselines, a six-condition synthetic corruption suite (fog, rain, low light, glare, blur and sensor noise at three severities), a learned-gate upper bound, and the gate ablation harness.

Every trainable component passes a small-scale verification gate before consuming any GPU time. All such gates are currently green: the variance head is confirmed to train stably, to produce non-degenerate variance, and — importantly — to increase its predicted uncertainty on deliberately corrupted inputs, which is the behaviour the method depends on. The evaluation metric suite was also pre-registered in code before results existed: detection expected-calibration-error, interval calibration and coverage, Gaussian negative log-likelihood, sparsification error, risk-coverage area, out-of-distribution AUROC and a self-contained COCO-style mAP implementation.

4.2 Dataset preprocessing — complete

The Pohang dataset has been fully prepared for training. Infrared frames were converted from 16-bit to 8-bit and letterboxed from 640×512 to 640×640 by padding only, with no rescaling, so that no geometric distortion is introduced; labels were transformed accordingly and out-of-bounds boxes clipped.

The most consequential correction was to the train/validation/test split. The split as originally delivered was frame-interleaved, which in 10 Hz video places near-identical consecutive frames on both sides of the split boundary and silently inflates every reported score. Both modalities were re-split into contiguous per-run blocks with guard bands between them, and split integrity is now verified mechanically before any training run: an audit checks for cross-split duplicates and for temporal proximity violations, and must exit clean. It currently passes on both modalities with zero duplicates and zero violations. The same ordinal cuts are applied to both modalities so that paired visible-infrared frames always land in the same split, which the fusion experiments require.

A visibility filter was additionally applied to the visible training stream only, removing boxes from the night run that carry no learnable signal in RGB while leaving validation and test data untouched. A native-resolution twin of the dataset was also built through hard links,

adding no new image bytes, so that higher-input-resolution experiments remain directly comparable to the 640×640 benchmark. An independent audit of the infrared pipeline was carried out and is documented: the 16-bit source carries approximately 7.7 effective bits, so 8-bit conversion loses essentially nothing, but the per-frame min-max normalisation used at conversion time inflates the dynamic range in about half of all frames because a single hot pixel keys the mapping. A percentile-clipped re-export is specified and costs only local CPU time.

Modality	Input used	Train	Validation	Test
Visible	640×640 (letterboxed from 2048×1080)	96,275	11,352	11,445
Infrared	640×640 (letterboxed from 640×512)	23,279	2,234	2,518

Table 1. Leakage-free split of the Pohang dataset after re-splitting and audit (frame counts).

4.3 Benchmarking — complete

The backbone benchmark has been executed on an NVIDIA H100 GPU server. Twenty-three model variants spanning five YOLO families and their size ladders were each trained with three independent seeds under an identical recipe, giving 68 completed training runs and approximately 390 GPU-hours of compute. Every row records its own software versions, configuration and a fingerprint of the exact data split used, so that no result can be confused with a run made under different conditions.

Rank	Variant	mAP@50-95 (mean ± SD)	mAP@50	Params (M)	GFLOPs
1	YOLOv9-e	0.3033 ± 0.0066	0.632	58.2	193.0
2	YOLOv9-c	0.3019 ± 0.0057	0.629	25.6	104.0
3	YOLOv8-x	0.2999 ± 0.0010	0.623	68.2	258.5
4	YOLO11-l	0.2994 ± 0.0041	0.623	25.4	87.6
5	YOLOv10-x	0.2982 ± 0.0037	0.625	31.8	171.8
...	YOLO11-n (smallest)	0.2486 ± 0.0028	0.578	2.6	6.6

Table 2. Backbone benchmark, visible modality, mean over three seeds (23 variants, 68 runs). Full results are recorded for all variants.

Three findings follow. First, the accuracy spread across the whole ladder is modest: the largest model beats the smallest by about 5.5 mAP points while costing roughly 22 times the parameters and 29 times the compute, so paying for size buys little on this data. Second, nine variants fall within one standard deviation of the best result and are therefore statistically indistinguishable, which is exactly why the tie-break rule was fixed in advance rather than chosen now. Third, applying that rule narrows the field to a short list of DFL-based, integration-friendly candidates — YOLOv9-c and YOLO11-l — that match the top score at under half the parameters of the nominal winner. The final pick is confirmed by the throughput measurement that opens the next phase.

A secondary observation carries directly into the scientific contribution: recall sits near 0.60 while precision sits near 0.82 across the entire ladder. The detector is not failing to be confident; it is missing objects, predominantly small and distant ones. That is precisely the regime in which a calibrated uncertainty estimate, and a second sensor weighted by it, are expected to help.

4.4 Reproducibility

The software environment is version-pinned (PyTorch 2.12.1 with CUDA 12.6, Ultralytics 8.4.90), all machine-specific paths are confined to a single configuration file, the work is committed at every phase boundary so that each result can be traced to the exact code that produced it, and every dataset modification is documented in a written change record. Benchmark runs are resume-safe and refuse to mix results produced under different data configurations.

5. Conclusion

The project has completed its planning, engineering and empirical-baseline stages. The architecture is designed and fully implemented, with every component verified on small-scale tests before consuming GPU time. The dataset is prepared, audited and free of the temporal leakage present in the original split, and its preprocessing is documented end to end. The benchmarking phase is finished: a 68-run, three-seed, 390 GPU-hour comparison across 23 model variants has established the deterministic baseline required by objective O1 and resolved the backbone question with a selection rule fixed before the results were seen. The measured error profile — high precision with limited recall on small, distant targets — supports the premise of the work, namely that reliability estimation and adaptive sensor fusion, rather than a larger detector, are where the remaining gains lie. What remains is the final training phase.

6. Future Scope

The remaining work is the final training phase, in which the implemented and already verified components are trained at full scale on the prepared data:

Uncertainty-aware training. Train the selected backbone with the Gaussian variance head, together with the parity check that confirms the added branch does not degrade detection accuracy relative to the deterministic baseline.

Baselines and calibration study. Train the Monte-Carlo Dropout and Deep Ensemble references and evaluate all methods against the pre-registered calibration metrics, on clean data and under the synthetic corruption suite.

Infrared confirmation and fusion. Train the short-listed variants on the infrared stream, compute the per-run infrared-to-visible transform from the dataset calibration, and evaluate the uncertainty-gated fusion system against single-modality and uncertainty-blind fusion, including the case where both sensors are degraded.

Cross-dataset validation. Complete annotation and onboarding of the MIT marine perception subset to test generalisation beyond a single sensor rig and geography.

Subject to these results, the intended outputs are a conference or journal submission reporting the calibrated single-pass uncertainty method and the uncertainty-gated fusion result, together with a public release of the evaluation harness.

A) List of Publications

SCI : Nil

Non-SCI : Nil (manuscript in preparation)

B) Conference Proceedings : Nil

C) Patents filed/Published/Granted : Nil



Signature of the Applicant



Signature of Mentor